PharmAchieve

SYNCOPE

Qualifying Exam Preparatory Course

COMPETENCIES COVERED IN THIS PRESENTATION...

Providing Care:
Clinical Care

Providing Care:
Distribution

Knowledge
& Expertise

Communication
& Collaboration

Leadership &
Stewardship

Professionalism

LEGEND

AV	Atrioventricular
BP	Blood Pressure
CV	Cardiovascular
ED	Emergency Department
ER	Emergency Room
OH	Orthostatic Hypotension
QOL	Quality Of Life
SCD	Sudden Cardiac Death
SSRI	Selective Serotonin Reuptake Inhibitor

DEFINITION AND PATHOPHYSIOLOGY

- A sudden and temporary loss of consciousness
  - Often referred to as "fainting"
- Majority of cases are benign, with patients having a quick and complete recovery
- Main complications arising from syncope are injuries from falling
- Some types of syncope include
  - Reflex syncope
  - Orthostatic hypotension syncope
  - Cardiac syncope

Reduction in venous blood flow to the heart (non-cardiac)

↓

Decreased cardiac output and blood pressure

↓

Impaired compensatory reflex response

↓

Cerebral hypoperfusion

↓

Syncope

RISK FACTORS

Family history

Age >70 years

Previous CV events

Emotional distress

Medical conditions

- CV disease
- Psychiatric conditions

Medications

- Antihypertensives
- Diuretics
- Antianginals

CLINICAL PRESENTATION

Prodromal presentation	Syncope	Post-syncope attack
• Dizziness • Lightheadedness • Fatigue/weakness • Visual disturbances • Nausea • Sweating/ flushing	• Loss of postural tone • Loss of consciousness • Rapid and self-limiting • Complete recovery of consciousness	• Threat of reoccurrence impairs QOL

DIAGNOSIS

Investigations*

- Obtain a clinical history and physical examination
- ECG to screen for cardiac syncope
- If indicated, carotid sinus massage to screen for carotid sinus hypersensitivity (particularly in those > 50 years of age)
- Measure BP after 5 minutes of lying down, then again after 1 and 3 minutes of standing to screen for OH

  → OH diagnosis is made when there is > 15 mmHg decrease in SBP or > 7 mmHg decrease in DBP
- Complete the Canadian Syncope Risk Score to risk-stratify patients in EDs – helps determine low-risk patients who can be safely discharged

* Investigations should be individualized for every patient

GOALS OF THERAPY

- ✓ Identify causes and eliminate where possible
- ✓ Prevent future recurrences
- ✓ Reduce mortality (cardiac syncope/falls-related)
- ✓ Minimize injury due to falls
- ✓ Avoid adverse effects of therapy

REFLEX SYNCOPE

Definition

- Most common type of syncope
- Body overreacts to a trigger and cannot compensate for loss of oxygen to brain.
- Triggers include severe pain, standing for too long, emotional distress
- Includes vasovagal syncope

Treatment

- Education and reassurance
- Avoid triggers
- Recognize prodromal symptoms
- Learn and practice isometric counter-pressure maneuvers for syncope prevention, e.g. crossing/stretching legs, tensing arms
- Increase fluid and salt intake (volume expansion)
  - Caution in kidney dysfunction and cardiac issues

PHARMACOLOGICAL MEASURES - REFLEX SYNCOPE

Class	Action	Comments
Alpha-1-adrenergic agonist midodrine	• Increases venous return	Clinical trials are lacking or have inconsistent evidence in this area
Beta-blockers	• May be more beneficial in those ≥42-years-old • Can increase bradycardia in carotid sinus syncope • Can increase risk of orthostatic hypotension	
Mineralocorticoid fludrocortisone	• Use if patient failed addition of simple salt supplements • Caution in patients with hypertension/heart failure	
SSRIs: paroxetine	• Uncommon	

ORTHOSTATIC HYPOTENSION SYNCOPE

Definition

- When moving from lying to sitting or sitting to standing
- May be caused by volume depletion, neurological disorders or drugs

Treatment

- Education and reassurance
- Avoid offending agents e.g. antihypertensives, antidepressants, diuretics, antianginals
- Recognize prodromal symptoms
- Learn and practice isometric counter-pressure maneuvers for syncope prevention, e.g. crossing/stretching legs, tensing arms
- Sleep with head elevated ~ 10° to prevent nocturnal hypertension
- Measure blood pressure -supine, sitting and upright
- Increase fluid and salt intake (volume expansion). Caution in kidney dysfunction and cardiac issues

PHARMACOLOGICAL MEASURES - ORTHOSTATIC HYPOTENSION SYNCOPE

Class	Effect
α_1-Adrenergic Agonists Midodrine	• Increases venous return
Mineralocorticoid fludrocortisone	• Use if patient failed addition of simple salt supplements • Fluid volume expansion secondary to renal Na+ retention and sensitizes peripheral alpha receptors • Caution in patients with hypertension/heart failure
Pyridostigmine	• Improves standing BP
Desmopressin	• Useful in patients with nocturnal polyuria
Erythropoietin	• Useful in anemic patients

Evidence is lacking for all drug options

CARDIAC SYNCOPE

Definition
• Various problems with heart cause syncope

Cause of cardiac syncope	Therapy
Sinus node dysfunction	• Cardiac pacemaker therapy
AV Block	• Cardiac pacemaker therapy
Paroxysmal supraventricular and ventricular tachycardias	• Catheter ablation and/or antiarrhythmic agents • Resolution of cause (i.e. ischemia)
Structural heart disease	• Implantable cardioverter defibrillator • Treat underlying disease

MONITORING PARAMETERS

EFFICACY ENDPOINTS			
Parameter	**Degree of Change**	**Frequency of monitoring**	**Timeframe**
# of episodes of syncope	Reduce	At every appointment	Duration of therapy
Orthostatic hypotension	Eliminate	Measure BP after patient has been supine for > 5 minutes then again after 1 and 3 minutes of standing MD: At initial visit and every appointment thereafter Patient: daily (BP diary)	
Falls	Prevent	Ongoing	

MONITORING PARAMETERS

SAFETY ENDPOINTS		
Parameter	**Degree of Change**	**Timeframe**
Supine hypertension	Prevent or minimize	Duration of therapy
Other adverse effects (edema, piloerection, pruritus)	Prevent or minimize	Duration of therapy

TAKEAWAY

Syncope is a sudden and temporary loss of consciousness

- It can be divided into three groups: reflex, orthostatic hypotension and cardiac

Treatment is targeted to the cause of syncope

- Reflex: inconsistent evidence for drug use
  - Patient education and assurance
  - Avoiding triggers of syncope
- Orthostatic hypotension: evidence for drug options is limited
  - Fludrocortisone and midodrine may be used
  - Patient education on slowly sitting up/standing
- Cardiac: treatment depends on cause
  - Examples include pacemaker, cardiac ablation, and internal defibrillators

CASE 1

TJ is a 76-year-old male presenting to the emergency department after a fall. His past medical history is significant for a previous myocardial infarction at age 72, hypertension, type 2 diabetes, hypercholesterolemia, and depression. His current medications include metoprolol 50mg BID, amlodipine 10mg daily, losartan 100mg daily, rosuvastatin 40mg daily, paroxetine 40mg daily, and vitamin D_3 4000 IU daily. He does not smoke or drink. TJ reports he fell after getting out of bed this morning. He states he gets dizzy a lot after standing up, but has never fallen due to this dizziness until today.

1. How many risk factors for syncope does TJ have?
   a) 6
   b) 5
   c) 4
   d) 3

2. Which of the following is the MOST likely reason for TJ's fall?
   a) Hypoglycemia
   b) Cardiac syncope
   c) Orthostatic hypotension syncope
   d) Reflex syncope

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3. Which of the following are agents that increase TJ's risk of syncope?
   a) Metoprolol
   b) Metformin
   c) Paroxetine
   d) Vitamin D3

4. Which of the following would you recommend to TJ to prevent another fall?
   a) Discontinuation of metoprolol
   b) Implantable defibrillator
   c) Cardiac pacemaker
   d) Slowly standing up from seated positions

REFERENCES

1. Solbiati M, Sheldon R. Cardiovascular Disorders: Syncope. *E-Therapeutics- Therapeutic Choices*. Canadian Pharmacists Association.
2. Guidelines for the diagnosis and management of syncope (version 2018). *European Heart Journal*;2018: 39, 1883-1948. doi:10.1093/eurheartj/ehy037.
3. Olshansky B. Pathogenesis and etiology of syncope. In: *UpToDate*.
4. Olshansky B. Reflex Syncope. In: *UpToDate*.
5. Longo DL, Fauci AS, Kasper DL, et al. Chapter 20. Syncope. *Harrison's Principles of Internal Medicine*.18 ed. McGraw- Hill; 2012.
6. Arnold, Amy C. et al. Orthostatic Hypotension: A Practical Approach to Investigation and Management. Canadian Journal of Cardiology, Volume 33, Issue 12, 1725 - 1728

CHANGE LOG

May 2025

- Formatting and grammar changes throughout
- Slide 8: Added note on vasovagal syncope
- Slide 9: Added beta- blockers can increase risk of hypostatic hypotension
- Slide 13: Expanded summary slide
- Slide 14-15: Added case

September 2025

- Formatting changes made
- Updated legend
- Slide 7: Added diagnosis
- Slides 14-15: Added monitoring parameters

February 2026

- Minor formatting edits, no change to content